

Senior Java Architect Interview Preparation (11+ Years) - Full Guide

Java 17/21

Q: What are Records?

A: Immutable data carriers used for DTOs. Example: record User(String name,int age).

Q: What are Virtual Threads?

A: Lightweight threads introduced in Java 21 for high concurrency systems.

Q: What is happens-before?

A: Ensures memory visibility across threads.

Q: HashMap vs ConcurrentHashMap?

A: ConcurrentHashMap is thread-safe; HashMap is not.

Spring Boot

Q: What is Spring Boot?

A: Framework for building production-ready apps quickly.

Q: What is Dependency Injection?

A: Inject dependencies via container instead of manual instantiation.

Q: What is @Transactional?

A: Manages transaction boundaries automatically.

Microservices

Q: What is Microservices Architecture?

A: Independent deployable services communicating over APIs.

Q: What is API Gateway?

A: Single entry for routing and security.

Q: What is Saga Pattern?

A: Handles distributed transactions across services.

Kafka

Q: What is Kafka?

A: Distributed event streaming system.

Q: What is Partition?

A: Unit for parallel processing in Kafka.

REST & SOAP

Q: What is REST?

A: Stateless API using HTTP.

Q: What is SOAP?

A: XML-based messaging protocol.

System Design

Q: Design a Payment System

A: Use API Gateway, Kafka, Payment Service, Fraud Service, DB with idempotency.

Advanced Question Set 1

Q: Explain architecture concept 1

A: Answer includes design, trade-offs, scalability, and production example for fintech systems.

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